THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The transport of soluble products of photosynthesis is called: 1

(a) transpiration **(b)** translocation **(c)** excretion **(d)** diffusion

2. Which part of the alimentary canal receives bile from the liver? 1

(a) large intestine **(b)** oesophagus **(c)** stomach **(d)** small intestine

3. The pump of the circulatory system is: 1

(a) kidney **(b)** heart **(c)** lungs **(d)** liver

4. The cells that control the opening of stomata are: 1

(a) xylem cells **(b)** mesophyll cells **(c)** guard cells **(d)** root hair cells

5. The opening and closing of the stomata is a function of: 1

(a) guard cells **(b)** mesophyll **(c)** epidermal cells **(d)** xylem

6. Which chamber of the heart pumps oxygenated blood to the body? 1

(a) right ventricle **(b)** right atrium **(c)** left ventricle **(d)** left atrium

7. Plants excrete waste products as: 1

(a) bile **(b)** urine

(c) sweat **(d)** gums, resins and latex

8. Which of the following is the main site of absorption of digested food? 1

(a) oesophagus **(b)** stomach **(c)** large intestine **(d)** small intestine

9. Valves are present in the heart to: 1

(a) mix the blood **(b)** prevent backflow of blood

(c) increase blood pressure **(d)** stop circulation

10. The pigment which absorbs light energy for photosynthesis is: 1

(a) chlorophyll **(b)** melanin **(c)** carotene only **(d)** haemoglobin

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the separation of oxygenated and deoxygenated blood useful in humans? 2

12. What is the function of digestive enzymes? 2

13. How is oxygen and carbon dioxide transported in human beings? 2

14. What are the methods used by plants to get rid of excretory products? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the double circulation of blood in human beings. 3

16. What is the importance of peristalsis, villi and intestine length in digestion? 3

17. What are the raw materials and products of photosynthesis? Write the equation. 3

OR

What are the differences between arteries and veins? 3

18. How does Amoeba obtain its food? 3

OR

Explain how fishes breathe. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the transport of water and minerals in plants with xylem, transpiration pull and root pressure. 5

OR

Explain the role of different digestive juices, enzymes and their sites of action with the final products. 5

20. Describe translocation in plants and compare it with transport in xylem. 5

OR

Why do plants and animals need different transport systems? Describe the human blood components and their functions. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND